

## DSCI551 HW4

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1. `db.countrylanguage.countDocuments({IsOfficial: "T", Percentage: {$gt: 0.5}})`

```
> db.countrylanguage.countDocuments({IsOfficial: "T", Percentage: {$gt: 0.5}})
< 202
```

2. `db.country.find({Continent: /America/, GNP: {$gt: 100000}},`  
`{_id: 0, Name: 1, GNP: 1})`

```
Name: 'Argentina',
GNP: 340238
}
{
  Name: 'Brazil',
  GNP: 776739
}
{
  Name: 'Canada',
  GNP: 598862
}
{
  Name: 'Colombia',
  GNP: 102896
}
{
  Name: 'Mexico',
  GNP: 414972
}
{
  Name: 'United States',
  GNP: 8510700
}
```

3. `db.country.aggregate([`

```
{ $match: { Continent: "North America", Population: { $gt: 100000 } } },  
{ $group: { _id: null, avgGNP: { $avg: "$GNP" } } },  
{ $project: { _id: 0, avgGNP: 1 } } })
```

```
> db.country.aggregate([  
  { $match: { Continent: "North America", Population: { $gt: 100000 } } },  
  { $group: { _id: null, avgGNP: { $avg: "$GNP" } } },  
  { $project: { _id: 0, avgGNP: 1 } } ])  
< {  
  avgGNP: 403444.625  
}
```

#### 4. db.country.aggregate([

```
{ $lookup: { from: "city", localField: "Capital", foreignField: "_id.ID", as:  
"cap_city" } }, { $unwind: "$cap_city" }, { $sort: { Name: 1 } }, { $limit: 5 },  
{ $project: { _id: 0, countryName: "$Name", cityName: "$cap_city.Name" } } ])
```

```
{  
  countryName: 'Afghanistan',  
  cityName: 'Kabul'  
}  
{  
  countryName: 'Albania',  
  cityName: 'Tirana'  
}  
{  
  countryName: 'Algeria',  
  cityName: 'Alger'  
}  
{  
  countryName: 'American Samoa',  
  cityName: 'Fagatogo'  
}  
{  
  countryName: 'Andorra',  
  cityName: 'Andorra la Vella'  
}
```

#### 5. db.countrylanguage.aggregate([

```
{ $group: { _id: "$_id.Language", count: { $count: {} } },  
{ $match: { count: { $gt: 20 } } }, { $sort: { count: -1, _id: 1 } } )
```

```
{  
  _id: 'English',  
  count: 60  
}  
{  
  _id: 'Arabic',  
  count: 33  
}  
{  
  _id: 'Spanish',  
  count: 28  
}  
{  
  _id: 'French',  
  count: 25  
}
```

```
6. db.city.aggregate([  
  { $match: { CountryCode: "USA" } },  
  { $group: { _id: "$District", avgPop: { $avg: "$Population" }, count: { $count: {} } },  
  { $match: { count: { $gt: 10 } } }, { $project: { avgPop: 1 } } ])
```

```
{  
  _id: 'Texas',  
  avgPop: 354164.6538461539  
}  
{  
  _id: 'California',  
  avgPop: 245833.91176470587  
}  
{  
  _id: 'Florida',  
  avgPop: 210093.86666666667  
}
```

```
7. db.city.insertMany([
  {"_id": {"ID": 4080 }, "Name": "LA suburb", "CountryCode": "USA", "District":
  "California", "Population": 1000},
  {"_id": {"ID": 4081 }, "Name": "LA dessert", "CountryCode": "USA", "District":
  "California", "Population": 2000}])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': {
      ID: 4080
    },
    '1': {
      ID: 4081
    }
  }
}
```

```
8. db.city.updateOne(
  {"_id.ID": 4081},
  {$set: {Name: "LA orange", Population: 3000}})
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
9. db.city.deleteMany({"_id.ID": {$in: [4080, 4081]}})
```

```
db.city.deleteMany({"_id.ID": {$in: [4080, 4081]}})
{
  acknowledged: true,
  deletedCount: 2
}
```

## Appendix (Results in MySQL):

This Appendix includes the results run in MySQL server for double-check.

| count(Language) |
|-----------------|
| 202             |

| Name          | GNP        |
|---------------|------------|
| Argentina     | 340238.00  |
| Brazil        | 776739.00  |
| Canada        | 598862.00  |
| Colombia      | 102896.00  |
| Mexico        | 414972.00  |
| United States | 8510700.00 |

| avg(GNP)      |
|---------------|
| 403444.625000 |

| Name           | Name             |
|----------------|------------------|
| Afghanistan    | Kabul            |
| Albania        | Tirana           |
| Algeria        | Alger            |
| American Samoa | Fagatogo         |
| Andorra        | Andorra la Vella |

| Language | count(*) |
|----------|----------|
| English  | 60       |
| Arabic   | 33       |
| Spanish  | 28       |
| French   | 25       |

| District   | avg(Population) |
|------------|-----------------|
| California | 245833.9118     |
| Texas      | 354164.6538     |
| Florida    | 210093.8667     |